MATRIX ACADEMY.....

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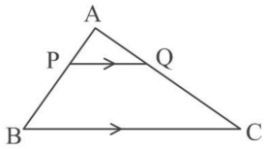
X – MATHEMATICS TEST

CHAPER – TRIANGLES

Time Allowed: 1 hour **Maximum Marks :** 30

Section A

1 In△ ABC, PQ ∥ BC . It is given that AP = 2.4 cm, PB = 3.6 cm and BC = 5.4 cm. PQ is equal to: [1]



A) 3.6 cm B) 2.16 cm C) 2.7 cm D) 1.8 cm

2 In△ ABC, D and E are points on side AB and AC respectively such that DE || BC and AD:DB=3:1. If EA = 3.3 cm, then AC = [1]

A) 5.5 cm B) 4.4 cm C) 4 cm D) 1.1 cm

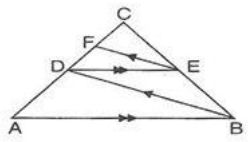
3 In△ ABC and, △ RPQ, AB = 4.5 cm, BC = 5 cm, CA = $6\sqrt{2}$ cm, PR = $12\sqrt{2}$ cm, PQ=10cm, QR=9 cm. If ∠A = 75° and ∠B = 55° , then ∠P is equal to [1]

A) 55° B) 75° C) 130° D) 50°

4 The sides of two similar triangles are in the ratio 4 : 7. The ratio of their perimeters is [1]

A) 16 : 49 B) 4 : 7 C) 12 : 21 D) 7 : 4

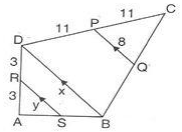
5 We have,AB ∥ DE and BD ∥ EF. Then, [1]



A) $AC^2 = BC\,.\,DC$ B) $AB^2 = AC\,.\,DE$

C) $BC^2 = AB\,.\,CE$ D) $DC^2 = CF.AC$

6 In the adjoining figuresRS ∥ DB ∥ PQ. If CP = PD = 11 and DR = RA = 3. Then, [1]

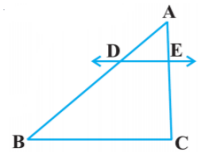


A) x = 14, y = 6. B) x = 10, y = 7. C) x = 16, y = 8. D) x = 12, y = 10.

For Question no, 7 and 8

A) Both A and R are true and R is the correct explanation of A.
B) Both A and R are true but R is not the correct explanation of A.
C) A is true but R is false.
D) A is false but R is true.

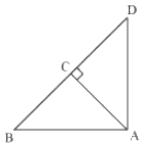
7 **Assertion (A):** D and E are points on the sides AB and AC respectively of a △ ABC such that AD = 5.7 cm, DB = 9.5 cm, AE = 4.8 cm and EC = 8 cm then DE is not parallel to BC. [1]
Reason (R): If a line divides any two sides of a triangle in the same ratio then it is parallel to the third side.



8 **Assertion (A):** $\triangle ABC \sim \triangle PQR$ such that $\angle A = 65°, \angle C = 60°$. Hence $\angle Q = 55°$. [1]
Reason (R): Sum of all angles of a triangle is $180°$.

Section B

9 $\triangle ABD$ is a right triangle right - angled at A and $AC \perp BD$. Show that $AB^2 = BC \times BD$ [2]



Section C

10 ABCD is a trapezium such that BC || AD and AB = 4 cm. If diagonals AC and BD intersect at O such that $\frac{AO}{OC} = \frac{DO}{OB} = \frac{1}{2}$. Then, Find DC. [3]

11 In$\triangle$ ABC, AD is a median and E is the midpoint of AD. If BE is produced, it meets AC at point F. Show that $AF = \frac{1}{3} AC$. [3]

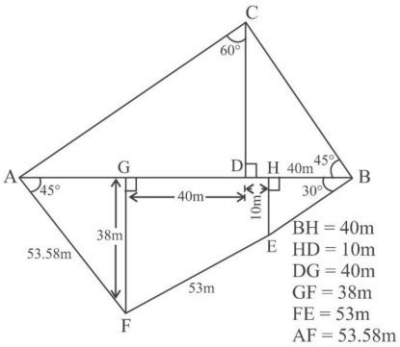
Section D

12 ABCD is a quadrilateral. P, Q, R and S are the points of trisection of sides AB, BC, CD and DA respectively and are adjacent to A and C.Prove that PQRS is a parallelogram. [5]

13 In$\triangle$ ABC , if $AD \perp BC$ and $AD^2 = BD \times DC$, then prove that $\angle BAC = 90°$. [5]

Section E

14 Mahesh and Ganesh are two friends whose plots are adjacent to each other. Mahesh's son made a drawing of the plots with necessary details.
It is decided that Mahesh will fence the triangular plot ABC and Ganesh will fence along the sides AF, FE and BE.
Observe the diagram carefully and answer the following questions:



(Use $\sqrt{2} = 1.41$ and $\sqrt{3} = 1.73$)

i. Find length BC. (1)

ii. Find length AG. (1)

iii. Calculate perimeter of$\triangle ABC$. (2)

OR

iv. Calculate length of $(AF + FE + EB)$. (2)